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Perceptions of Artificial Intelligence and Its Impact on Academic Integrity Among University Students in Peru and Chile: An Approach to Sustainable Education

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Abstract: In a context where artificial intelligence (AI) is transforming higher education, this study analyzes how students' perceptions of AI influence their academic integrity (INA), with a focus on sustainable education. Through a correlational-explanatory analysis based on Structural Equation Models (SEMs) applied to a sample of 659 students from 13 universities in Chile and Peru, it is observed that AI has a significant and direct impact on academic integrity in both countries ($\beta = 0.44$). In Peru, the most influential dimension is trust in education ($\lambda = 0.86$), followed by social, economic, security, and risk implications ($\lambda = 0.78$), while attitudes towards AI also have a direct impact on integrity factors ($\beta = 0.15$). In Chile, the dimensions of trust in education ($\lambda = 0.83$) and social and economic impact ($\lambda = 0.79$) are most relevant, and the relationships between the dimensions of academic integrity such as justice, respect, and responsibility ($\lambda = 0.71$) are stronger. The study highlights the importance of incorporating AI literacy into educational curricula and developing regulatory frameworks that promote its ethical use, linking these actions to sustainable education. The findings highlight the need for sustainable educational approaches that enhance understanding of AI and ensure that its use in academia is beneficial, ethical, and contributes to sustainable development.

Keywords: artificial intelligence; academic integrity; students; sustainable education; SDG4; quality education



Citation: Espinoza Vidaurre, S.M.; Velásquez Rodríguez, N.C.; Gambetta Quelopana, R.L.; Martinez Valdivia, A.N.; Leo Rossi, E.A.; Nolasco-Mamani, M.A. Perceptions of Artificial Intelligence and Its Impact on Academic Integrity Among University Students in Peru and Chile: An Approach to Sustainable Education. *Sustainability* **2024**, *16*, 9005. <https://doi.org/10.3390/su16209005>

Academic Editors: Jesús N. García and Hao-Chiang Koong Lin

Received: 31 July 2024

Revised: 12 October 2024

Accepted: 15 October 2024

Published: 17 October 2024



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1. Introduction

Artificial intelligence (AI) has undergone significant development in recent years, leading to its application in various disciplines, particularly in education. These AI systems can be trained through machine and deep learning to mimic the functions of the human brain, allowing them to perform routine tasks through the analysis and processing of large amounts of data [1–3]. AI is characterized by computer systems that are able to replicate human cognitive processes such as learning, adaptation, synthesis, and self-correction, using data to perform highly complex information processing tasks [4,5].

Led by the United Nations, global efforts to achieve significant progress in all countries have focused on “promoting quality education” (Sustainable Development Goal 4) to reduce existing gaps and create equitable and inclusive opportunities, in line with the principles of sustainable education [6]. In this context, artificial intelligence (AI)-based tools have become an integral part of everyday life, including education. These tools not only help to optimize and facilitate academic activities but also promote continuous and adaptive learning, which is essential for sustainable education in an ever-changing world [7].

In higher education, educators and students alike are embracing and benefiting from the many advantages of artificial intelligence. However, concerns regarding how academic integrity and current understandings of disciplinary knowledge might be challenged are being raised [8]. The introduction of AI-based applications also raises a number of ethical considerations and challenges that need to be carefully assessed and managed. These implications cover a broad spectrum, including issues related to privacy, bias, academic integrity, and accountability in decision-making. This landscape underscores the importance of rigorous analysis and prudent management when implementing these technologies in education, ensuring that their use is consistent with the values and ethical principles of the academic community [2,9].

One of the most important aspects that a university must promote is academic integrity. Maintaining this integrity is essential and must be the central axis of the university's educational and administrative activities. It is a fundamental educational responsibility that institutions must address directly. Therefore, academic integrity must be integrated into all issues related to education and must not be ignored or treated in isolation [10]. Academic integrity is essential for educational institutions to develop individuals who are committed to their core values and who can be agents of positive change in their communities. Therefore, the primary purpose of these higher education institutions should not only be academic success but also to inculcate core values such as honesty, trust, justice, respect, responsibility, and courage, as proposed by the International Centre for Academic Integrity (ICAI) [11]. These values are expected to play a fundamental role in developing ethical and responsible leaders for future generations.

To justify the need for this study, it is essential to recognize that, despite the exponential growth in the use of artificial intelligence in education, there is a significant gap in the literature in terms of an in-depth understanding of how students' perceptions towards these technologies affect academic integrity. The existing literature has focused primarily on the technical applications of AI and its pedagogical benefits but has left the ethical and behavioral implications that these tools may have in the academic environment relatively unexplored. This study seeks to fill this gap through providing empirical evidence that can inform educational policies and pedagogical practices to promote the responsible and ethical use of AI in higher education institutions. In doing so, it not only contributes to the existing literature but also provides concrete tools to address current challenges related to the education of future professionals.

The main objective of this study is to analyze the influence of artificial intelligence (AI) on Academic Integrity (AIN), comparing the differences between university students from Peru and Chile. To this end, a Structural Equation Model (SEM) was used to identify how different dimensions of AI, such as attitudes towards AI, understanding and knowledge, attitudes towards teaching, trust in education, ethics and responsibility, social and economic impact, and security and risk, influence students' AIN. In addition, the variations and relationships between these dimensions are explored, taking into account the educational contexts of both countries.

The analysis involved the collection of several dimensions selected from a sound theoretical framework. The selection of these dimensions was based on their relevance and support in the academic literature, as well as widely accepted theories in the field of study [12].

Therefore, this study set out to answer the following research question: What is the influence of students' perceptions of artificial intelligence (AI) on their Academic Integrity? The hypothesis of this study is that students' perceptions of AI have a significant impact on their academic integrity. This approach responds to the need to investigate how the socio-educational contexts of both countries modulate students' perceptions of the use of AI tools and how these interactions affect their academic integrity behavior.

The research questions and context justify the need for this study, given the exponentially increasing use of AI tools in higher education and the impact this has on the education and ethics of their use by students. This study helps to fill a gap in the existing literature

through providing empirical evidence that can contribute to future research and the formulation of better policies that promote the responsible use of AI in university environments.

The results of this research allowed us to differentiate and deepen the characteristics of AI in relation to academic integrity—a topic of recent importance in the field of global education. Unlike previous studies, this one focuses on a different socio-educational geographical context, namely, Peru and Chile, which share a border. In this sense, the results explore—according to different authors—the AI factors that are common and that influence the perceptions and behaviors of university students regarding the progress of AI and its relationship with academic integrity. The comparative results allowed us to identify patterns as well as significant differences between the two countries, which will allow us to design more resilient educational policies aimed at improving mechanisms in academic training.

1.1. Literature Review

1.1.1. Artificial Intelligence (AI) in Education

Technology, a central pillar of the 21st century, plays a fundamental role in various areas of daily life, including education [13]. Recent technological innovations have created significant opportunities for teaching and learning methods through providing tools to enhance these processes. The adoption of active teaching methods and the generation and dissemination of knowledge have been strengthened with the development of these new technologies. The growing importance of technology in education is reflected in the integration of digital skills in the academic curricula of several countries and the constant encouragement of teachers to implement these tools in their teaching environments [14].

In the field of emerging technologies, artificial intelligence (AI) is a particularly noteworthy innovation that offers the possibility of establishing novel and effective models in the evaluation process within higher education institutions [15]. AI can be conceptualized as a set of computer systems designed to emulate essential human capabilities, such as learning, adaptability, synthesis, and self-correction. These systems use data to perform highly complex processing tasks [16]. Their integration into the educational process should be carried out in a significant way, incorporating materials and software with capabilities such as abstract reasoning, autonomous learning, adaptability to new situations, and advanced interaction, thus emulating the characteristics of intelligent entities [17,18]. The implementation of these features, together with various active learning approaches, has gained significant interest in the educational field, leading to a progressive increase in research involving the use of artificial intelligence in this area [19].

In addition, it is essential to integrate specific training in AI into academic curricula, beyond programs that focus exclusively on computer science. This will prepare students to successfully meet the demands of the labor market in their future careers [5,20].

Recent advances in the artificial intelligence (AI) field have increased the complexity and challenges associated with related technologies. Chatbots, for example, have been greatly improved in their ability to produce text in a near-human way, complementing their capacity to handle large amounts of information. This development poses new challenges for both teachers and students, who have to recognize the existence of tools capable of generating information that is not always accurate or reliable. However, beyond these challenges, it is important to emphasize that AI also offers potential benefits, such as supporting teaching tasks and enhancing learning. Therefore, if used appropriately and responsibly, AI can be a valuable tool [21–25].

The dimensions of the AI variable adapted by the authors [5,26–30] include attitudes towards AI, understanding and knowledge, attitudes towards teaching it, and confidence in training with AI. They also cover aspects of ethics and responsibility, social and economic impacts, and safety and risk considerations. These dimensions provide a comprehensive framework for analyzing the influence and adoption of AI in different contexts.

In [29,31], the increasing importance of artificial intelligence (AI) and machine learning (ML) algorithms in adaptive learning has been examined, with a particular focus on the

need to develop explainable AI. These studies highlighted the importance of designing AI systems that provide clear and understandable explanations for decision making and recommendation generation, which is essential for building trust and ensuring the ethical use of these technologies in education. They also analyzed the tools and methods used in these studies and identified significant limitations, such as a lack of transparency in algorithms and insufficient attention to the reasoning underlying adaptive learning outcomes. These shortcomings underline the relevance of the present study, which aims to contribute to the development of more explainable and accessible AI systems for both students and educators, thus promoting a more ethical and responsible e-learning environment.

1.1.2. Academic Integrity

The integration of artificial intelligence (AI) raises several concerns for those involved in education [7], and the accelerating development and increasing sophistication of AI tools is raising significant concerns in the academic community, particularly regarding their potential to undermine standards of originality in scholarly production. This issue focuses on how AI can be used to circumvent the principles of originality in thinking and writing, which are key elements in the traditional understanding of academic endeavors. While these concerns are valid in the context of the traditions of higher education, they also reveal and perpetuate a culture that privileges a particular mode of knowledge representation through written text. Disproportionately valuing writing over other forms of creative expression promotes a goal of techno-solutionism that seeks to preserve the integrity of the writing process, which faces significant challenges in the age of AI [32].

Discussions about the incorporation of artificial intelligence (AI) in education have focused primarily on ethical issues. Ref. [33] argues that while AI systems can expand access to information, concerns about academic integrity existed prior to their implementation, and AI does not significantly alter these pre-existing dynamics. For their part, Ref. [34] proposes an ethical and collaborative integration of generative AI (GenAI) systems as a way to support the development of 21st century skills.

Faced with the growing threat that AI poses to academic integrity, the higher education sector has responded in a variety of ways, characterized by their fragmentation. While some have moved quickly to impose outright bans on the use of AI, others have opted for a strategy of adaptation, developing and disseminating guidelines for students on how to interact effectively and ethically with these technologies. This diversity of responses underlines the complexity of integrating technological innovation into the ethical and pedagogical frameworks of educational institutions [35].

The main concerns in the academic context arise from the ability of AI to produce academic texts, raising questions about the validity and integrity of assessment processes. Addressing these challenges will require reconsideration of how the knowledge and skills of students are assessed, replacing traditional approaches with more innovative strategies that AI cannot perform effectively. Another critical aspect is the increasing reliance on digital technologies in education, in particular the excessive use of AI applications to complete tasks and solve problems. This trend could lead to a decline in human cognitive abilities and create relevant deficits in students' academic and professional development [36].

The successful adoption of technological innovations depends, to a large extent, on user acceptance. Despite the benefits and limitations of digital technologies, their effective integration into learning processes is inextricably linked to the attitudes of educators and learners towards their use for educational purposes [37].

Dimensions of the Academic Integrity variable, adapted from the following authors: [38–41], include honesty, trust, justice, respect, responsibility, and factors that influence integrity. These dimensions provide a framework for understanding ethical behavior in academia.

This study not only contributes to the understanding of the impact of artificial intelligence on academic integrity but also offers a comprehensive approach that advocates for the responsible implementation of AI in education, in line with the principles of sustainability.

By proposing an ethical and pedagogical framework that mitigates the risks associated with the use of AI, the study reinforces the need for sustainable education, where technology not only enhances learning but does so in an equitable, inclusive, and responsible manner. Developing more transparent and explainable AI systems will not only build trust among students and educators but also ensure that the integration of these technologies is done in an ethical manner, promoting a lasting and sustainable learning environment that meets the demands of the 21st century.

2. Methods

2.1. Design

In this study, a quantitative approach was used to analyze the relationship between artificial intelligence (AI) and academic integrity (AIN) among university students in Peru and Chile. For this purpose, Structural Equation Modeling (SEM) was used as the main analytical technique, as this technique allows the synchronous examination of direct and indirect relationships between latent and observed variables [42,43]. Structural modeling is an appropriate technique for this type of study since it allows for the structural modeling of the complex interactions between the different dimensions that make up the study variables.

The study was conducted using surveys to systematically collect data using a correlational-explanatory design. This approach seeks to identify the relationship between two study variables and to explain, through a model, the influence of one variable on the other [44–46].

2.2. Participants

The study included 659 students selected by simple random probability sampling, which ensures the representativeness of the sample. Inclusion criteria were enrolment in the current semester and voluntary informed consent. Those who did not meet these criteria were excluded. Data collection was carried out through online questionnaires distributed by email and institutional WhatsApp groups to ensure ethical participation.

The questionnaire was distributed to 1000 university students in Chile and Peru, achieving a response rate of 65.9% ($n = 659$). Data collection took place between 8 April and 24 May 2024. In terms of gender distribution, the overall sample consisted of 45.2% males and 54.8% females. In Chile, male participation was predominant (56.9%), while in Peru the proportion of women was higher (62.1%). Table 1 shows the distribution of respondents by country and gender. Statistical analysis using the Student t test showed that these differences were statistically significant ($p < 0.001$), suggesting different demographic and cultural influences between the two countries.

Table 1. Sex ratio (%) of students surveyed in Chile and Peru, 2024.

Gender	Chile	Perú	Frequency
Male	56.9	37.9	45.2
Female	43.1	62.1	54.8
Total number	100.0	100.0	100.0

Note. Indicates the gender ratio of students between Peru and Chile.

2.3. Study Instrument

The formulation of the theoretical framework constituted the initial phase of the present study, as it was fundamental to the development of the conceptual framework that addresses the factors of artificial intelligence that affect the academic integrity of university students in Peru and Chile, as shown in Table 2.

Table 2. Research tool.

Variable	Dimensions	References
Artificial Intelligence	Attitudes towards AI; Understanding and knowledge of AI; Attitudes towards teaching AI	Almaraz-López et al. (2023) and Sit et al. (2020) [5,30].
	Confidence in training with AI	Almaraz-López et al. (2023), Pacheco-Mendoza et al. (2023), Sit et al. (2020), Incio & Capuñay (2023), Hamoud et al. (2018), and Mensah et al. (2023) [5,26–30].
	Ethics and responsibility; Social impact Economic impact and Safety and risk	Pacheco-Mendoza et al. (2023), Incio & Capuñay (2023), Hamoud et al. (2018), and Mensah et al. (2023) [26–29].
Academic Integrity	Honesty	Henning et al. (2020), Cutri et al. (2021), and Araya et al. (2023) [39–41].
	Trust	Cutri et al. (2021) y Araya et al. (2023) [39,40].
	Justicia	Farrelly & Baker (2023) and Araya et al. (2023) [38,40].
	Respect, Responsibility Factors influencing integrity	Cutri et al. (2021) y Araya et al. (2023) [39,40]. Cutri et al. (2021) y Farrelly y Baker (2023) [38,39].

Note. Based on a review of the literature and prepared by the authors.

The research instrument was articulated through two different questionnaires specifically designed to assess the variables of interest, namely, academic integrity and artificial intelligence. The first section covered informed consent, followed by general questions about the participants. These questions included demographic data such as gender, academic information related to their major, academic year, type of university, and age.

The first questionnaire focused on assessment of the academic integrity variable and was structured around six fundamental aspects and consisted of 71 items designed to examine the following factors: honesty, trust, justice, respect, responsibility, and other factors that influence academic integrity. This instrument used a five-point Likert scale to record the participants' responses, with a value of 1 indicating "strongly disagree", 2 "disagree", 3 "neutral", 4 "agree", and 5 "strongly agree".

The second questionnaire was organized into a total of 23 items focused on assessing students' attitudes towards various aspects of artificial intelligence (AI), such as understanding and knowledge of the topic, attitudes towards its teaching, confidence in training with AI, ethics and responsibility, social impact, economic impact, and perceptions of safety and risk. The total number of questions (considering both questionnaires) was 94. In addition, at the end of the survey, participants were offered the option of receiving the results of the study by email, which was done with those who gave their consent.

The third section used a dichotomous approach to determine whether respondents had received AI training. If so, additional information was requested on the type, duration, and perceived quality of such training.

Prior to administering the main questionnaire, a pilot test was conducted with 39 undergraduate students from universities in Tacna, Peru, to assess the reliability of the instrument. The internal consistency coefficients obtained ranged from 0.89 to 0.92, indicating high reliability. In the main sample ($n = 659$), reliability coefficients of 0.934 were observed for the AI dimension and 0.943 for academic integrity, confirming the robustness of the instrument. Subsequently, three PhD experts validated the content of the questionnaire by rating aspects such as clarity, objectivity, and relevance on a scale from 0 to 30 points. Scores between 28 and 30 reflected the appropriateness of the questionnaire for the purpose of the study.

2.4. Data Collection Procedure

The research materials were delivered to the research offices of each university studied. They were given all the details about the study and asked to help distribute the questionnaire. Participants who showed interest in virtual collaboration were invited to fill out an online form implemented using Google Forms. This form was distributed to students at the universities included in the study sample, using email and the WhatsApp platform as means of communication. Initially, 1000 questionnaires were distributed in order to obtain responses several times higher than the calculated sample size, and a total of 659 responses were received.

Participation in the research was strictly voluntary, with a firm commitment to protecting the confidentiality of participants' personal information. In order to avoid the collection of identifying information, the questionnaire focused on questions aimed at demographic characterization related to the university institutions involved. The questionnaire was available during the months of April and May 2024, closing access at the end of this period.

2.5. Statistical Analysis

Data analysis in this study was carried out using IBM SPSS v.29.0 and SPSS AMOS v.26.0 software. These tools were used to carry out Exploratory Factor Analysis (EFA), Confirmatory Factor Analysis (CFA), and Structural Equation Modeling (SEM). In the CFA and in the final SEM model, different goodness-of-fit tests were applied, following the recommendations of [47,48] Byrne (2016), who suggests evaluating the CFI, RMSEA, GFI, and TLI indices. These indices made it possible to confirm the validity of the structural models, provided that the data showed a good fit in the context of Peruvian and Chilean students.

Through the structural models for both groups of students, Peruvian and Chilean, and in accordance with the objective of the study, the two SEM models presented complete interactions that allowed the simultaneous evaluation of the influence of AI on academic integrity.

3. Results

3.1. Empirical Results

Student Profile

Regarding the average age of the students surveyed, it was 22 years (24 for Chile and 21 for Peru). A higher percentage of respondents for the Peruvian case was observed in the age cohort of 16 to 21 years (69.50%) while, in Chile, the same proportion of students answered the survey (30.00%) at different ages (16–21 years, 22–25 years, and 26–29 years).

The Peruvian students surveyed stated that they studied at the Universidad Andina del Cusco (5.46%), Catholic Sedes Sapientiae (10.77%), National of Moquegua (10.93%), National Hermilio Valdizan (2.43%), National Jorge Basadre Grohmann (17.75%), and Private of Tacna (15.17%). As for Chilean students, they indicated studying at Arturo Prat University (9.86%), Antofagasta University (0.30%), Atacama (0.15%), Bio-Bio (0.15%), La Serena (0.46%), Santo Tomás University (15.63%), and Inacap (10.93%). The survey was answered in the different years of study, but with a higher percentage of students up to the third year (66.30%).

Regarding the academic program studied, it can be observed from Table 3 that, for both countries, the highest percentage of respondents was from the professional engineering and architecture program (46.60% in Chile and 22.20% in Peru), followed by 22.31% from business sciences (23.40% in Peru and 20.60% from Chile), 20.33% from health sciences (23.30% from Chile and 18.50% from Peru), and, to a lesser extent, from education (10.47%), law (7.59%), history (6.37%), and others (1.37%).

Table 3. Academic program studied by students from Chile and Peru, 2024.

Academic Programme That Includes the Study of	Chile	Peru	Total
Health Sciences	23.30	18.50	20.33
Business sciences	20.60	23.40	22.31
Engineering and Architecture	46.60	22.20	31.56
Educational sciences	6.70	12.80	10.47
Law	0.40	12.10	7.59
History	0.00	10.30	6.37
Other	2.40	0.70	1.37
Total number	100.00	100.00	100.00

Note. The results are based on the survey used.

3.2. Empirical Model

3.2.1. The First Theoretical Model

Following a literature review, this study proposes a theoretical model to explore the influence of artificial intelligence (AI) on academic integrity (AIN) of university students in Peru and Chile. The model suggests that different dimensions of AI, including attitudes towards AI, understanding and knowledge, attitudes towards teaching, trust in AI training, ethics and responsibility, social impact, economic impact, security, and risk, affect academic integrity. Academic integrity, in turn, is analyzed through dimensions such as honesty, trust, justice, respect, responsibility, and factors that influence integrity. This structural model allows us to examine how these interrelated dimensions affect the ethical behavior of university students.

3.2.2. Exploratory Factor Analysis (EFA)

Table 4 presents the results of the Exploratory Factor Analysis (EFA) performed on the items related to the dimensions of the artificial intelligence (AI) variable, revealing five distinct factors. As shown in Table 5, these factors together account for 64.108% of the total variance. The first factor, comprising eight items, explains 43.541% of the variance, indicating a strong association with items related to confidence in training with AI. The remaining factors include the dimensions of social impact, economic impact, safety and risk, understanding and knowledge, attitudes towards AI, and ethics and responsibility.

Table 4. Exploratory factor analysis of artificial intelligence.

	Confidence in Training with AI (CT)	Social Impact, Economic Impact and Satefy and Risk (SE)	Understanding and Knowledge of AI (UK)	Attitudes Towards AI (AT)	Ethics and Responsibility (ER)
Item 82. By the end of my training, I will be confident in using basic AI tools to develop academic work where appropriate.	0.878				
Item 83. By the end of my training, I'll have a better understanding of AI performance evaluation methods for academic work.	0.858				
Item 84. Overall, I believe that at the end of my training I will have more knowledge to work regularly with AI in my academic work.	0.765				
Item 81. All students should be trained in artificial intelligence.	0.762				
Item 79. Teaching AI will help my career.	0.551				
Item 80. I feel comfortable using AI.	0.532				

Table 4. Cont.

	Confidence in Training with AI (CT)	Social Impact, Economic Impact and Satefy and Risk (SE)	Understanding and Knowledge of AI (UK)	Attitudes Towards AI (AT)	Ethics and Responsibility (ER)
Item 85. AI will play an important role in academic education.	0.520				
Item 86. My knowledge allows me to use AI efficiently.	0.326				
Item 90. Measures are needed to ensure ethical and rights-respecting use of artificial intelligence.		0.905			
Item 91. Ethical principles and AI regulations should be promoted in my field of study.		0.812			
Item 93. It is important to evaluate the benefits associated with the use of artificial intelligence in my field of study.		0.805			
Item 94. AI has optimized processes and improved task efficiency where implemented.		0.601			
Item 92. AI systems respect my autonomy and give me control over decisions that directly affect me.		0.498			
Item 77. I understand the limitations of artificial intelligence.			0.815		
Item 75. I understand and/or know the basic computational principles of AI.			0.706		
Item 76. I am familiar with the terminology associated with artificial intelligence.			0.593		
Item 73. I am LESS likely to consider a career or speciality as AI advances.				0.859	
Item 74. Some professions and specialities will be replaced or substituted by AI during my lifetime.				0.504	
Item 87. Artificial intelligence helps people progress and become more efficient.					0.632
Item 88. The results and decisions generated by artificial intelligence systems are easy to understand and explain.					0.485

Note. The table presents the results of the Exploratory Factor Analysis (EFA) of the variable artificial intelligence, identifying five main factors obtained using IBM SPSS v.29.0 software.

Table 5. Factorial matrix and explained variance of artificial intelligence dimensions.

	CT	SE	UK	AT	ER
Explained variance (%)	43.541	7.382	5.589	4.278	3.318
Cumulative variance (%)	43.541	50.923	56.512	60.79	64.108
Confidence in training with AI (CT)	1.000				
Social impact, Economic impact and Satefy and risk (SE)	0.710	1.000			
Understanding and knowledge of AI (UK)	0.632	0.563	1.000		
Attitudes Towards AI (AT)	0.285	0.204	0.443	1.000	
Ethics and responsibility (ER)	0.680	0.675	0.630	0.331	1.000

Note. Shows factorial matrix and explained variance for AI dimensions.

Table 6 presents the results of the Exploratory Factor Analysis (EFA) applied to the items related to the dimensions of the Academic Integrity Variable (AIN), showing the existence of four distinct factors. Table 7 shows these four factors, which explain 56.319% of the total variance. The first factor, composed of 18 items, explains 29.242% of the variance, suggesting a strong association between these items and the justice, respect, and responsibility dimension. The other dimensions include factors related to the influence of integrity, honesty, and trust.

Table 6. Exploratory factor analysis of academic integrity.

	Fairness, Respect and Responsibility (FR)	Factors Influencing Integrity (FI)	Honesty (HO)	Trust (TR)
Item 28. In an academic community, respect is defined as the recognition of the value of its members in their individual and collective character.	0.826			
Item 27. An academic community should foster respect among students, faculty and staff. It should also promote respect for scholarship, research, educational processes and its intellectual heritage.	0.805			
Item 34. Members of an academic community have a responsibility to act with integrity in all their activities and not to stand idly by in the face of injustice or dishonesty.	0.799			
Item 33. An academic community must uphold the highest standards of conduct in learning, teaching and research, and requires all its members to act responsibly to promote academic integrity.	0.791			
Item 29. The teaching and learning process, as well as working together, depends on mutual respect.	0.786			
Item 35. Shared responsibility empowers everyone to make a difference, combats apathy and reinforces the value of each individual within the academic community.	0.768			
Item 30. Pupils show respect by being punctual, attentive, receptive, prepared, participative and timely with their work.	0.755			
Item 25. Students expect the University to handle academic dishonesty fairly to protect teaching quality and degree value.	0.750			
Item 31. Show respect by avoiding personal attacks, offensive language, bullying, unnecessary requests for reassessment, and disruptive behaviour in class or faculty interactions.	0.742			
Item 24. Students expect the tools and criteria used to assess their work to be accurate, fair and relevant.	0.731			
Item 26. Students must make responsible use of the sources they consult and cite them appropriately.	0.727			
Item 32. The academic community shows respect by crediting and properly citing others' work across all formats.	0.721			
Item 22. Fairness is essential in any evaluation process.	0.715			
Item 36. In sharing responsibility for maintaining standards of academic integrity, one of the most difficult issues is how to deal with the dishonesty of others.	0.679			
Item 23. Without fairness, ratings can be false, misleading and arbitrary.	0.666			
Item 40. A system of academic integrity should require individuals to take responsibility for their own honesty and to seek to prevent misconduct by others.	0.566			
Item 20. An academic community should ensure standards and practices for academic integrity and interactions are founded on justice.	0.505			
Item 21. Pupils and teachers are constantly evaluating their ideas, data and work.	0.502			

Table 6. Cont.

	Fairness, Respect and Responsibility (FR)	Factors Influencing Integrity (FI)	Honesty (HO)	Trust (TR)
Item 50. Personal insecurity influences the commission of breaches of academic integrity.		0.802		
Item 49. Poor time management affects academic integrity.		0.786		
Item 51. The habit of doing things at the last minute influences the commission of academic integrity offences.		0.766		
Item 52. Not knowing how to do a job or task influences the commission of academic integrity offences.		0.742		
Item 48. The fear of failing a course encourages people to commit breaches of academic integrity.		0.727		
Item 43. Pressure from family members to complete their studies early influences the commission of academic integrity offences.		0.715		
Item 44. Family expectations of student competence influence the commission of academic integrity offences.		0.677		
Item 46. Low socio-economic status influences the commission of academic integrity offences.		0.674		
Item 47. An affluent socio-economic situation influences the commission of breaches of academic integrity.		0.669		
Item 42. The possibility of being eliminated from the race influences the commission of academic integrity offences.		0.629		
Item 53. The ease with which work can be copied rather than produced influences the commission of academic integrity offences.		0.625		
Item 41. Fear of losing one's scholarship influences the commission of academic integrity offences.		0.623		
Item 45. The ease with which information can be found on the Internet influences the commission of academic integrity offences.		0.601		
Item 60. The amount of work to be submitted in a short period of time influences the risk of committing a breach of academic integrity.		0.513		
Item 16. When administrators interact with students in a respectful and responsible manner, they build their confidence.			0.774	
Item 12. A climate of trust is created in the classroom to encourage the free exchange of ideas between students.			0.763	
Item 13. At university and in the classroom, actions and policies that promote and justify the trust of others are encouraged.			0.762	
Item 14. Teachers set clear guidelines for academic work and its assessment, thereby building students' confidence.			0.760	
Item 15. Students do their work honestly and encourage teachers to pay more attention to them and to engage in open academic dialogue, even when it takes them down unexpected paths.			0.733	
Item 17. Mistrust impoverishes academic life.			0.569	
Item 18. Without trust, members of a university community work in isolation.			0.561	
Item 19. Without trust, there can be no free exchange of ideas.			0.531	
Item 06. Taking an exam for someone else, or having someone else take an exam for you.				0.789
Item 07. Use of hidden notes in written exams.				0.766

Table 6. *Cont.*

	Fairness, Respect and Responsibility (FR)	Factors Influencing Integrity (FI)	Honesty (HO)	Trust (TR)
Item 04. Resubmitting an assignment submitted in one subject for assessment in another subject.				0.727
Item 05. Buy an academic paper.				0.721
Item 03. Copying from another student during an examination without that student being aware of it.				0.684
Item 02. Changing the words of someone else's material and presenting it as your own.				0.667
Item 08. Cut and paste from the web or smart applications with website recognition into the bibliography.				0.583

Note. The table shows the results of the Exploratory Factor Analysis (EFA) for the academic integrity variable, identifying four main factors obtained using IBM SPSS v.29.0 software.

Table 7. Factorial matrix and explained variance of the dimensions of academic integrity.

	FR	FI	HO	TR
Explained variance (%)	29.242	14.339	7.897	4.841
Cumulative variance (%)	29.242	43.580	51.478	56.319
Fairness, Respect and Responsibility (FR)	1.000			
Factors influencing integrity (FI)	0.313	1.000		
Honesty (HO)	0.614	0.158	1.000	
Trust (TR)	−0.188	0.192	−0.043	1.000

Note. Shows the factor matrix and explained variance for the AIN dimensions.

3.2.3. Confirmatory Factor Analysis (CFA)

Table 8 presents the descriptive statistics for each dimension of the study variables, highlighting the variability in the responses, with standard deviation values ranging from 0.78 to 0.90. The negative asymmetry observed in several dimensions suggests a widespread positive perception among students. Most of the dimensions show adequate internal consistency, with α coefficients higher than 0.70. However, the AT dimension presents an α of 0.62, which, although lower than the recommended threshold, maintains a Composite Reliability (CR) and Average Variance Extracted (AVE) above the reference values, justifying its inclusion. Similarly, the FI dimension has an AVE of 0.45 but is retained in the model due to its high CR (0.92) and α (0.93) values. These dimensions are retained because of their relevance in the exploratory context of the study of artificial intelligence and because they contribute significantly to the understanding of the influence of AI on academic integrity. Finally, it is proposed to evaluate these dimensions in a Confirmatory Factor Analysis (CFA) in order to deepen the analysis, provided that they do not negatively affect the overall indices of the model.

The Confirmatory Factor Analysis (CFA) performed on the dimensions of artificial intelligence (AI) validated the theoretical structure of the proposed model, confirming that AI is composed of five dimensions: Trust in Training (CT); Social, Economic, Security and Risk Impact (SE); Understanding and Knowledge (UK); Attitudes towards AI (AT); and Ethics and Responsibility (ER). The obtained goodness of fit indices reflect a good fit of the model with values of CFI = 0.951 and GFI = 0.923, both above the recommended thresholds. Similarly, the value of TLI = 0.941 confirms the robustness of the model, while an RMSEA of 0.059, lower than the threshold of 0.08, indicates that the theoretical model is consistent with the observed data (see Figure 1).

Table 8. Descriptors for the dimensions of the study variables.

Dimensions	SD	A	α	CR	AVE
Confidence in training with AI (CT)	0.79	−0.45	0.91	0.91	0.56
Social impact, Economic impact and Satefy and risk (SE)	0.83	−0.60	0.89	0.88	0.61
Understanding and knowledge of AI (UK)	0.85	−0.24	0.78	0.78	0.54
Attitudes Towards AI (AT)	0.90	−0.41	0.62	0.71	0.58
Ethics and responsibility (ER)	0.85	−0.49	0.75	0.75	0.60
Fairness, Respect and Responsibility (FR)	0.79	−0.87	0.95	0.95	0.52
Factors influencing integrity (FI)	0.83	−0.61	0.93	0.92	0.45
Honesty (HO)	0.78	0.58	0.87	0.88	0.50
Trust (TR)	0.86	−0.83	0.90	0.89	0.50

Note. The table presents the descriptive statistics for each dimension of the study variables, including standard deviation (SD), skewness (A), Cronbach's alpha (α), Composite Reliability (CR), and Average Variance Extracted (AVE).

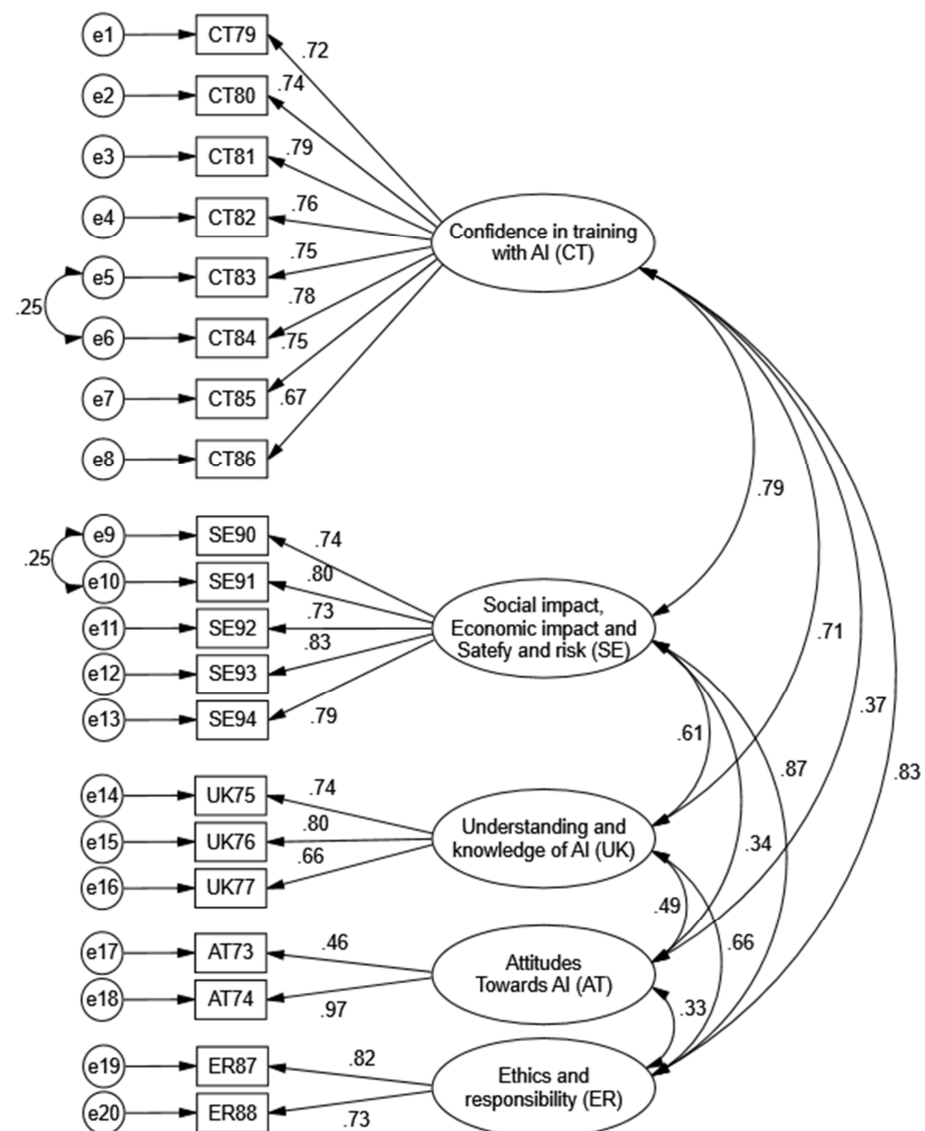


Figure 1. The coefficients of each item reflect the intensity of its relationship with the corresponding factor, supporting the validity of the factorial structure of the model. Confirmatory Factor Analysis (CFA) of the dimensions of AI, carried out with SPSS AMOS v.26.0.

The Confirmatory Factor Analysis (CFA) carried out on the dimensions of the variable Academic Integrity (INA) confirmed that the theoretical model is structured in four dimensions: Justice, Respect and Responsibility (FR), Factors Influencing Integrity (FI), Honesty (HO) and Trust (TR). The fit indices obtained for this model show a good fit, with values of CFI = 0.961, TLI = 0.955, and GFI = 0.902, all of which are above the recommended threshold of 0.90. In addition, the RMSEA of 0.035, which is well below the accepted threshold, indicates that the model is consistent and adequately fits the observed data (see Figure 2).

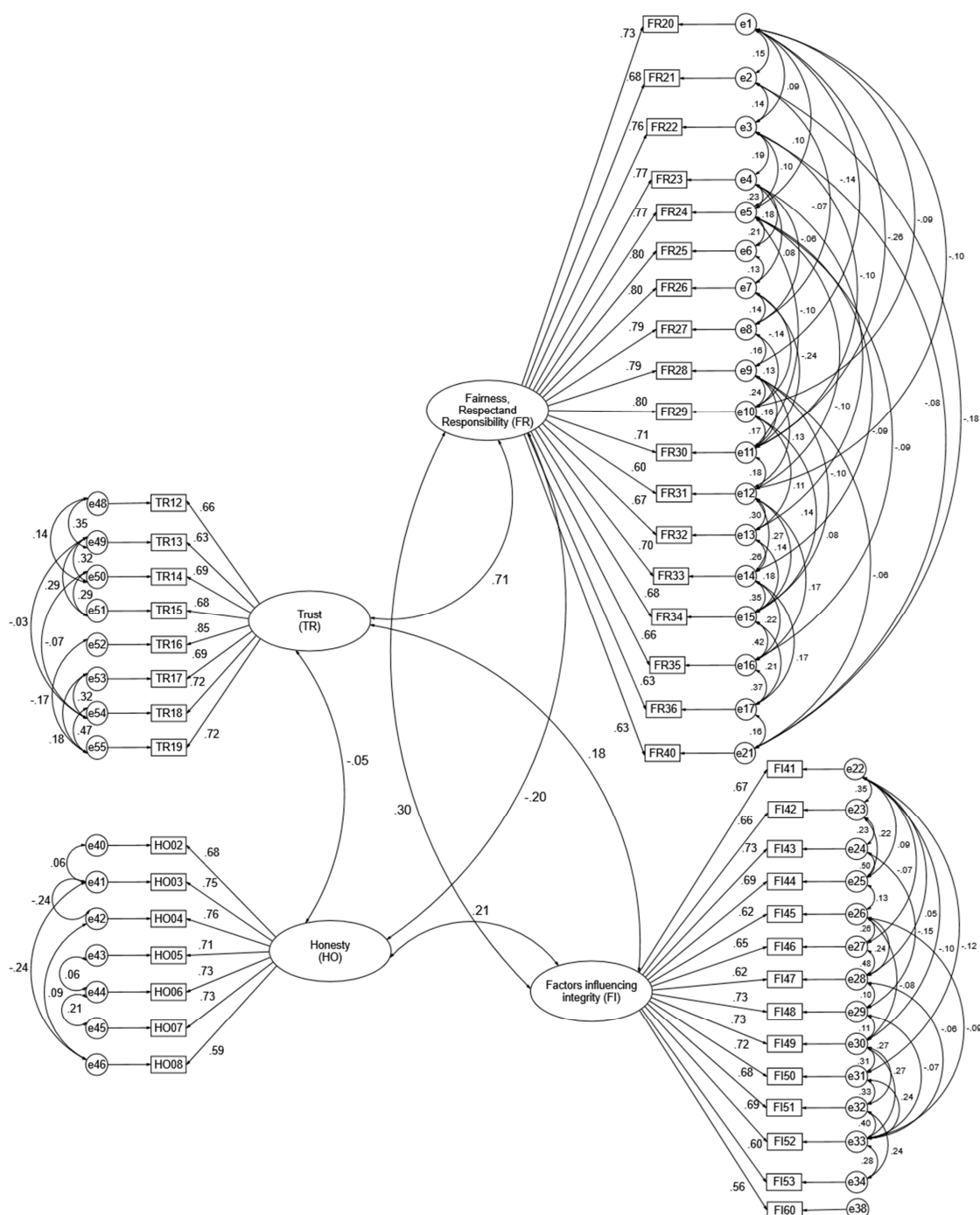


Figure 2. The coefficients of each item reflect the intensity of its relationship with the corresponding factor, supporting the validity of the factorial structure of the model. AFC–Dimensions of Academic Integrity, carried out with AMOS of SPSS v.26.0.

3.2.4. Final Structural Model

Initially, a conceptual model was proposed in which artificial intelligence and academic integrity were structured into different dimensions based on the literature review. However, after applying Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA), the theoretical model was refined, consolidating a final Structural Model (SEM). This SEM model allows a more precise assessment of the influence of AI on academic integrity among university students in Peru and Chile, providing a more comprehensive understanding of the impact of AI on ethical behavior based on the cultural context. The results of the final SEM model confirm the significant impact of AI on academic integrity and explain the differences observed between the two countries.

3.2.5. The SEM Model—Peru

The structural model for Peruvian students shows that AI has a direct and significant impact on academic integrity, with a value of $\beta = 0.44$, indicating a positive causal relationship. The most influential dimension of AI is Trust in Education (CT), with a factor loading of $\lambda = 0.86$, followed by the Social, Economic, Security and Risk (SE) dimension with $\lambda = 0.78$. The other dimensions also make significant contributions to the model.

An important result is the direct and positive influence of the attitude towards AI on the factors of academic integrity, with a value of $\beta = 0.15$, indicating that the attitude of Peruvian students towards AI has a significant impact on the factors that determine academic integrity.

The model fit indices are satisfactory, with CFI = 0.963, GFI = 0.951, and RMSEA = 0.077, indicating a good fit. Furthermore, the TLI of 0.944 supports the robustness of the SEM model, confirming its structural validity (see Figure 3).

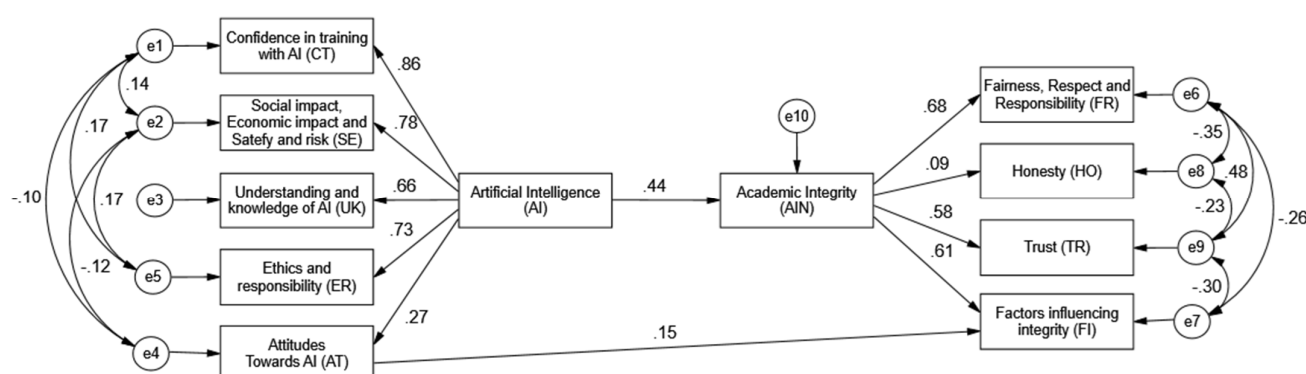


Figure 3. SEM model representing the relationships between the dimensions of artificial intelligence (AI) and Academic Integrity (AIN) in university students in Peru, developed by AMOS of SPSS v.26.0.

3.2.6. The SEM Model—Chile

The structural model for Chilean university students shows that the influence of artificial intelligence (AI) on academic integrity (INA) is comparable to that observed in the Peruvian case, with a value of $\beta = 0.44$. The AI dimensions with the highest factorial weight are trust in education ($\lambda = 0.83$) and social, economic, security, and risk impact (SE) ($\lambda = 0.79$), which stand out as the most influential. Regarding academic integrity, the relationships between its dimensions are stronger than in the Peruvian model; specifically, the justice, respect, and responsibility (FR) dimension has a factorial weight of $\lambda = 0.71$, suggesting that Chilean students perceive a stronger link between these dimensions and AI.

Furthermore, the goodness of fit indices for the Chilean students' model are superior, with values of GFI = 0.966, CFI = 0.993, and RMSEA = 0.035, indicating an excellent fit. The structural robustness is also confirmed by a TLI of 0.989, which reinforces the validity of the model (see Figure 4).

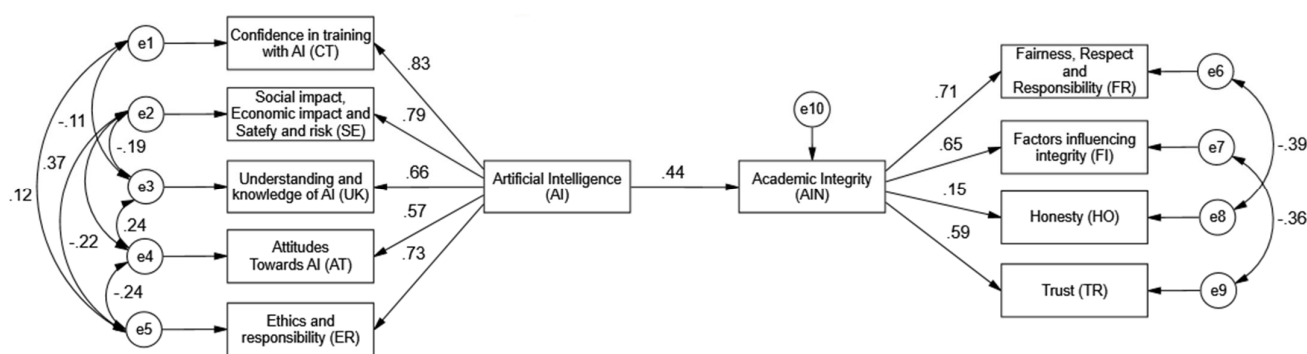


Figure 4. SEM model representing the relationships between the dimensions of artificial intelligence (AI) and Academic Integrity (AIN) in university students in Chile, developed by AMOS of SPSS v.26.0.

Figures 3 and 4 provide a structural analysis that directly answers the research question about the influence of artificial intelligence (AI) on students' Academic Integrity (AIN). In both models, the coefficient $\beta = 0.44$ indicates a positive and significant causal relationship between AI and AIN, suggesting that as positive perceptions of AI increase, so does students' academic integrity. This value supports the hypothesis that perceptions of AI have a direct impact on AIN, which is consistent in the Peruvian and Chilean contexts. Thus, the empirical results presented in the figures confirm that the dimensions of AI are related to the practice of academic integrity values, confirming the central hypothesis of the study.

3.2.7. Explained Variability

The SEM models applied to both Peruvian and Chilean students show an explained variance of $R^2 = 0.20$ in both contexts, indicating that these models are able to explain 20% of the variance in the influence of AI on academic integrity, based on the dimensions included in the final structural model. This result suggests that the remaining 80% of the variance is influenced by factors not included in the model, which could be related to dimensions such as educational policies, social and family environment, culture, among others.

4. Discussion

The SEM analysis for the cases of Peru and Chile reveals a significant influence of AI on students' academic integrity in both countries. In the structural model considered, the structure of the factorial weights for the AI dimensions is similar in both contexts. However, in the Chilean case, the dimensions related to academic integrity show stronger relationships, with better factorial loadings. In addition, the global fit indices of the SEM model are higher in the Chilean context, suggesting the presence of cultural and educational differences between the countries.

An important finding of the Peruvian SEM model, which is not observed in the Chilean model, is the influence of attitudes towards AI on factors related to academic integrity. This suggests that, for Peruvian students, positions on the role of AI have a more significant impact on their ethical behavior and decisions related to academic integrity. In contrast, this relationship does not seem to be as influential for Chilean students.

Although the explained variance is similar in both countries, suggesting that the influence of AI on academic integrity is comparable, there are differences in how some specific dimensions affect academic integrity. In particular, attitudes towards AI play a more determinant role in the Peruvian context, which may reflect differences in the adoption and use of AI in the educational environments of both countries.

The results obtained answer the research question posed: What is the influence of students' perceptions towards artificial intelligence (AI) on their Academic Integrity (AIN)? The general hypothesis is confirmed, showing that perceptions towards AI have a significant influence on students' academic integrity, with differences depending on the educational context in Peru and Chile.

During the COVID-19 pandemic—which had a significant impact on the economy and prompted a shift to distance learning modalities in many educational institutions—faculty members adopted advanced technologies for teaching. At the same time, universities intensified their efforts to strengthen their technological capabilities for a post-pandemic scenario. This increased use of technology in education has facilitated a more widespread adoption of artificial intelligence (AI) in the classroom [6,49,50].

The introduction of artificial intelligence (AI) significantly increases the complexity and scope of current technological challenges. In light of this situation, it is crucial that academic and professional communities address essential questions related to defining the role that human intelligence should play in relation to AI, identifying effective methods for integrating both intelligences, analyzing the possibilities for collaboration and mutual enrichment between human and artificial intelligence, and determining the new skills and knowledge that should be developed and promoted, as has been suggested by the Organization for Economic Co-operation and Development (OECD) [51].

Comparing the results of this study with those reported by [29], there was a convergence in the recognition of the growing relevance of artificial intelligence (AI) in education. This convergence highlights how the expansion of learning needs contributes to the greater role of AI in education. In this sense, the results of students by country reflect different realities. While the impact of AI on integrity was perceived only in terms of security and risk by Chilean students, Peruvian students tended to consider the impacts of both the economic impact and security and risk.

These findings highlight the importance of a contextualized approach when addressing academic integrity in different countries. The variability in the importance of the factors between Chile and Peru could be related to cultural, socio-economic, and educational differences that influence the perception and evaluation of academic integrity. This finding highlights the need to design educational strategies and policies that take into account these contextual particularities in order to effectively promote academic integrity through adapting to the specific realities and needs of each country.

This analysis reinforces the idea that educational interventions and policies must be context-sensitive, recognizing that the same factors may have different effects depending on the socio-cultural and economic environment in which students find themselves.

Comparing the findings of [52] with those of the present study highlights the importance of addressing the discrepancy between negative perceptions about the impact of AI on academic integrity and positive opinions about its ability to enhance students' confidence in the learning process. These findings underscore the need to integrate sustainable approaches to education that not only mitigate ethical concerns about the use of AI but also promote the responsible and beneficial use of this technology in educational settings. This approach to sustainable education allows technological advances to be balanced with practices that promote long-term ethical and academic development, benefiting both students and educational institutions in their commitment to integrity and lifelong learning.

The findings of [36] indicate that the implementation of artificial intelligence (AI) in the field of education poses various challenges for educators, and the need to adapt pedagogical techniques and methods to leverage the benefits of recent advances in AI and mitigate the associated difficulties was highlighted. This finding emphasizes the importance of overcoming these barriers to ensure the effective management of AI in a manner that is consistent with the principles of academic integrity. In this sense, our findings allow us to define the lines of action as challenges for educators working on issues that involve AI, including identification of the risks and safety of its use.

Comparing the results of [38] with those obtained in this study, the significant impact of artificial intelligence (AI) on the academic integrity of students was confirmed. Both studies suggest that AI has the potential to positively impact the educational environment through providing students with access to assistive technologies that can enhance their learning and academic success. Given the complex implications of AI in education, it is critical to focus on the ethical use of these technologies, promote AI literacy, and establish

regulatory frameworks that empower both students and educators to safely and effectively harness the potential of AI.

The results obtained in this research indicate that AI has a highly significant impact on students' academic integrity. These results are consistent with the findings of [35,53–55] who stated that there is a significant risk that students may misuse AI models to generate plagiarized content or cheat on assignments and assessments. This concern is one of the most frequently cited in the contemporary academic literature and has led to an ongoing debate about whether these technologies should be banned outright in higher education institutions (HEIs).

Plagiarism and academic misconduct are persistent problems in higher education that have been the subject of extensive research over time. These dishonest practices pose a significant threat to academic integrity and have been extensively analyzed in the professional literature [56]. The introduction of artificial intelligence (AI) has amplified these threats for two main reasons: the lack of clear academic policies and the difficulty in detecting its misuse. A recent study found a notable lack of specificity in academic policies regarding the use of AI technologies. Out of a total of 142 higher education institutions (HEIs) surveyed in May 2023, only one institution had implemented an explicit ban on the use of AI [57].

Future Directions and Research Opportunities

As artificial intelligence (AI) algorithms become more prominent in the learning field, the need for explainable AI has become increasingly apparent. This growing demand stems from the desire of educators and students to understand the decision-making processes and recommendations generated by these algorithms. Future research will focus on developing AI systems that provide clear and transparent explanations of how they work, which is expected to make it easier for users to understand the logic behind the results of adaptive learning. The main purpose of explainable AI in this context is to provide students and educators with accessible and understandable explanations of the processes of algorithms, thus fostering trust, responsibility, and the ethical use of AI in e-learning environments. Explainable AI aims to be a key tool in clarifying the methods and reasons behind certain recommendations made by AI models [31,58].

5. Conclusions

The study shows a significant impact of artificial intelligence (AI) on academic integrity among university students in Peru and Chile, with an impact coefficient of $\beta = 0.44$. This finding highlights the role of AI in shaping ethical practices and behaviors, which raises the need to carefully evaluate its integration in the educational field in order to promote strong ethical values. The analyzed dimensions of AI—including attitudes, understanding, ethics, and responsibility—emerge as key aspects that influence the perception and application of academic integrity in these contexts, reinforcing the importance of ethical adoption of AI in university settings.

From a comparative perspective, the SEM analysis reveals both similarities and differences between the Peruvian and Chilean contexts. In the Peruvian case, it was observed that attitudes towards AI significantly influence factors related to academic integrity, highlighting a direct correlation between the perception of AI and ethical values, represented by a coefficient of $\beta = 0.15$. This aspect is not presented in the same way in the Chilean structural model, which could reflect cultural differences regarding technology and ethics in each country. However, the Chilean model presents a statistically better fit, suggesting that the relationship between AI and academic integrity is more coherent and structured in this context.

Another important difference is that in Chile the relationships between the dimensions of academic integrity are stronger than in Peru, especially the dimension of Justice, Respect, and Responsibility (FR), which has a higher factorial weight ($\lambda = 0.71$). This suggests that Chilean students have a clearer perception and stronger connection between these

dimensions and AI, which strengthens the coherence of the relationship between AI and ethical behavior. This variability in the links suggests that attitudes towards AI and its integration into academic culture may be more developed in the Chilean context, which could explain the greater consistency of the model in this country.

At the level of fit and explained variance, both SEM models show that the influence of AI on academic integrity explains 20% of the variance in the data, which means that 80% of the variance remains unexplained. This percentage suggests that there are other contextual and cultural factors, such as educational policy, family and social environment, and institutional culture, that may also contribute to the relationship between AI and academic integrity. The inclusion of these additional variables could provide a more comprehensive understanding of how AI affects ethical behavior in university settings.

The findings of this study have relevant implications for higher education, particularly with regard to the integration of advanced technologies such as AI into academic training within a sustainable education framework. It is important not only to harness the technological benefits of AI but also to understand how it affects the ethical and behavioral dimensions of students to ensure that its adoption promotes responsible and sustainable practices in the long term. Higher education institutions need to develop educational policies that not only promote the use of AI but also ensure its ethical implementation, especially in areas such as training in ethical values, individual and social responsibility, and respect for sustainability principles. In this way, higher education can play an essential role in preparing future professionals capable of using AI responsibly and consciously, thus contributing to a more ethical and sustainable development in academia and beyond.

The teacher plays an important role in teaching technology, artificial intelligence, and its healthy use, as it generates very interesting skills in the development of the student, forming him or her as an integral professional who can face the world of work. This new role of the teacher goes hand in hand with sustainable education, combined with the contribution of target 4.7 of SDG 4 on quality education.

Limitations of the Study

A limitation of the study was the variability of participation between countries. Differences in representativeness and response rate were observed between students from Chile and Peru, which may have biased the comparability of results between the two countries. This variability may limit the generalizability of the findings and suggests the need to consider these factors in future research to obtain a more balanced and comparable representation of both populations.

Another limitation was that of self-report bias. The methodology used was based on self-administered questionnaires, which means that participants' responses were subject to possible social desirability bias. Despite the efforts made to minimize this bias, it is possible that participants may have provided answers that were in line with what they considered to be socially acceptable, which could affect the validity of the data obtained.

Author Contributions: Conceptualization, R.L.G.Q. and A.N.M.V.; methodology, S.M.E.V. and N.C.V.R.; software, M.A.N.-M., E.A.L.R. and N.C.V.R.; validation, S.M.E.V., R.L.G.Q. and A.N.M.V.; formal analysis, A.N.M.V., S.M.E.V., R.L.G.Q., N.C.V.R. and M.A.N.-M.; investigation, N.C.V.R., S.M.E.V. and M.A.N.-M.; data curation, N.C.V.R., S.M.E.V. and M.A.N.-M.; writing—original draft preparation, S.M.E.V. and R.L.G.Q.; writing—review and editing, E.A.L.R. and S.M.E.V.; visualization, A.N.M.V. and R.L.G.Q.; supervision, N.C.V.R.; project administration, S.M.E.V. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Universidad Privada de Tacna, Grant RESOLUTION No. 411-2023-UPT-CU, and the APC was funded by the Universidad Privada de Tacna.

Institutional Review Board Statement: The study was conducted in accordance with the tenets of the Declaration of Helsinki, ensuring that the inclusion of individuals was based solely on their voluntary consent and that the privacy of their personal data was rigorously protected. In addition, it was verified that the students were able to complete the informed consent form independently

and on their own initiative, which facilitated meaningful participation while respecting the ethical principles required in research. The study was approved by the Ethics Committee, in accordance with Law No. 003-2024 CEI/ESPG-UPT, and validated by the University through the Vice Rector for Research, which convened an Institutional Review Board composed of blind peers. As a result, RESOLUTION No. 411-2023-UPT-CU was issued on 20 November 2023, authorizing the study to be carried out in accordance with current regulations.

Informed Consent Statement: Informed consent was obtained from all subjects included in the study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Acknowledgments: The REDES research group would like to thank the Universidad Privada de Tacna for its support, especially the Graduate School.

Conflicts of Interest: The authors declare no conflicts of interest.

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